torch.squeeze#

https://pytorch.org/docs/stable/generated/torch.squeeze.html#torch.squeeze

Description:

Returns a tensor with all specified dimensions of input of size 1 removed. For example, if input is of shape: (A×1×B×C×1×D)(A \times 1 \times B \times C \times 1 \times D)(A×1×B×C×1×D) then the input.squeeze() will be of shape: (A×B×C×D)(A \times B \times C \times D)(A×B×C×D). When dim is given, a squeeze operation is done only in the given dimension(s). If input is of shape: (A×1×B)(A \times 1 \times B)(A×1×B), squeeze(input, 0) leaves the tensor unchanged, but squeeze(input, 1) will squeeze the tensor to the shape (A×B)(A \times B)(A×B). The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

Function Signature

```
torch.squeeze(input: Tensor, dim: Optional[Union[int,
List[int]]]) → Tensor#

Parameters

if given, the input will be squeezed
```

Code Examples

```
>>> x = torch.zeros(2, 1, 2, 1, 2)
>>> x.size()
torch.Size([2, 1, 2, 1, 2])
>>> y = torch.squeeze(x)
>>> y.size()
torch.Size([2, 2, 2])
>>> y = torch.squeeze(x, 0)
>>> y.size()
torch.Size([2, 1, 2, 1, 2])
>>> y = torch.squeeze(x, 1)
>>> y.size()
torch.Size([2, 2, 1, 2])
>>> y = torch.squeeze(x, (1, 2, 3))
torch.Size([2, 2, 2])
```

Notes & Warnings

NOTE

Note The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

⚠ **WARNING**

Warning If the tensor has a batch dimension of size 1, then squeeze(input) will also remove the batch dimension, which can lead to unexpected errors. Consider specifying only the dims you wish to be squeezed.

Detailed Documentation

Documentation

Returns a tensor with all specified dimensions of input of size 1 removed.

For example, if input is of shape: (A×1×B×C×1×D)(A \times 1 \times B \times C \times 1 \times D)(A×1×B×C×1×D) then the input.squeeze() will be of shape: (A×B×C×D)(A \times B \times C \times D)(A×B×C×D).

When dim is given, a squeeze operation is done only in the given dimension(s). If input is of shape: (A×1×B)(A \times 1 \times B)(A×1×B), squeeze(input, 0) leaves the tensor unchanged, but squeeze(input, 1) will squeeze the tensor to the shape (A×B)(A \times B)(A×B).

The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

If the tensor has a batch dimension of size 1, then squeeze(input) will also remove the batch dimension, which can lead to unexpected errors. Consider specifying only the dims you wish to be squeezed.

input (Tensor) – the input tensor.

dim (int or tuple of ints, optional) –

only in the specified dimensions.

Changed in version 2.0: dim now accepts tuples of dimensions.